

Problem Set 4: Asymmetric Information, Moral Hazard & Adverse Selection

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [date — before Lecture 1 of Module 5]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with $\langle AI \rangle$ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (Piece rates, flat wages, and selling the job — 15 points). *A worker produces output worth \$1 per unit; output equals effort e . Effort costs the worker $c(e) = e^2/20$ dollars and is unobservable — only output can be contracted on. The worker's outside option is 0.*

- (a) **(4 pts)** *Under a piece rate r per unit, the worker maximizes $re - e^2/20$. Show that she chooses $e = 10r$, and compute her effort, earnings (gross pay), effort cost, and net payoff at $r = 0.4$.*
- (b) **(3 pts)** *Under a flat wage (paid regardless of output), what effort does a purely self-interested worker choose, and why? One sentence on what this predicts about flat-wage workplaces — a prediction Lecture 2's gift-exchange evidence complicates.*
- (c) **(5 pts)** *The firm keeps output minus pay, so under piece rate r its profit is $e - re = 10r(1 - r)$. Find the firm's profit-maximizing r , the resulting effort, firm profit, and worker net payoff. Then show that “selling the job” — piece rate $r = 1$ plus a fixed franchise fee $F = 5$ — achieves the first-best effort $e = 10$ and makes the firm strictly better off. Hint: first-best effort maximizes total surplus $e - e^2/20$.*
- (d) **(3 pts)** *Now output is noisy (equal to effort on average) and the worker is risk-averse. In words plus one inequality: why does the sell-the-job contract stop being optimal, and what does the optimal piece rate trade off?*

Question 2 (Participation and incentive constraints — 15 points). *A project yields revenue \$120 on success and \$30 on failure. With high effort (personal cost \$12 to the agent) success probability is 0.9; with low effort (cost \$0) it is 0.5. The agent's outside option is \$20. Effort is invisible; the contract is a wage pair (w_s, w_f) paid on the outcome. Both parties are risk-neutral; wages may not be negative.*

- (a) **(4 pts)** *Write the participation constraint and the incentive constraint — each first in one plain-English sentence, then in one line of algebra. Simplify the incentive constraint to a statement about the spread $w_s - w_f$.*

- (b) **(5 pts)** Find the cheapest contract implementing high effort (both constraints bind). Report (w_s, w_f) , the expected wage, and verify that implementing high effort beats the firm’s best flat-wage-low-effort alternative.
- (c) **(3 pts)** Explain in two sentences why no spread larger than \$30 helps the firm — and why, with any risk aversion at all on the agent’s side, a larger spread strictly hurts. What economic role does each binding constraint play?
- (d) **(3 pts)** If the agent were risk-averse, state (no computation) which constraint becomes more expensive to satisfy, what happens to the firm’s cost of implementing high effort, and when the firm would give up and accept low effort.

Part B: Experimental Design (45 points)

Question 3 (Decomposing the performance-pay effect). Lazear (2000) found Safelite’s switch to piece rates raised output $\sim 44\%$, roughly half from incentives (the same workers producing more) and half from selection (more productive workers joining). Design an experiment — lab, online, or field — that separately identifies the incentive and selection effects of a piece-rate contract for a real task. Specify:

- (a) **(6 pts)** Subject pool, recruitment, and the real-effort task, with the properties the task must have (measurable output, a quality margin, no team production) and why each matters.
- (b) **(12 pts)** Treatments. Your design must include (i) a stage or arm where workers are randomly assigned to fixed wage vs. piece rate, and (ii) a stage or arm where workers choose their contract before working. Say precisely which comparison identifies the incentive effect, which identifies selection, and what the contract-choice data themselves reveal.
- (c) **(9 pts)** Measurement: output quantity AND quality (piece rates famously distort unmeasured margins), plus a baseline productivity measure taken before any contract is revealed — explain what the baseline buys you for the selection analysis.
- (d) **(9 pts)** Hypotheses and rough sample-size reasoning: anchor effect sizes on Lazear’s decomposition (total $\sim 44\%$, roughly half incentives), state the per-arm n needed to distinguish the incentive effect from zero and from the total effect. Simulation-based reasoning is welcome; formulas are not required.
- (e) **(9 pts)** The biggest confound in YOUR design and your answer to it.

Part C: Silicon Pilot Interpretation ⟨AI⟩ (25 points)

Question 4 (The agents that matched both literatures). You run `module4-simulation-lab.ipynb` with neutral personas. Your silicon workers deliver effort ≈ 5 under the \$2.50 piece rate (the exact selfish-benchmark optimum), effort ≈ 2 under the stingy flat wage, and effort ≈ 7 under the generous flat wage (tight distributions — within-cell standard deviations under 1) — a textbook incentive response AND a strong gift-exchange pattern, simultaneously. Your lab partner celebrates: “we’ve validated standard theory and Fehr et al. in one run!”

- (a) **(8 pts)** Explain why producing BOTH canonical patterns at once is itself grounds for suspicion rather than double validation, and give two mechanisms (at least one LLM-specific) that could generate this result.

- (b) **(9 pts)** Design the diagnostic simulation that determines what is driving the flat-wage effort pattern. Your design must include (i) a parametric wage sweep (many wage levels between \$10 and \$25, identical wording) and (ii) an intentions manipulation (wage set by a computer vs. chosen by the employer), and may add others (e.g. stripping the experiment framing). State the predicted result pattern under each mechanism from part (a). Run it if API budget allows and report; otherwise state the predictions precisely.
- (c) **(8 pts)** A colleague wants to use this lab to pilot a real gift-exchange field experiment — including choosing the generous wage level and predicting the effort response. State one thing the silicon pilot CAN safely inform, one thing it CANNOT (and why), and the design change this implies for the human study.

Attach your AI transcripts and, if you ran code, the results CSV.